Jesús M. Rueda-Becerril, Ph.D.

Seattle, WA | jm.ruebe@gmail.com | +1 (765) 430-2330 | linkedin.com/in/jeruebe | altjerue.github.io

Professional Summary

Computational physicist (Ph.D.) with 10+ years writing scientific code, running computational experiments, and validating models against data. Currently evaluates frontier language models on expert-level physics reasoning and builds neural-network surrogates of expensive simulations. Author of open-source scientific libraries (Paramo, Tleco; ApJ 2024) with code review and unit testing, and mentor of graduate students.

Skills

Languages: Python, C/C++, Fortran 95, R, Rust, Octave, SQL, Java, Julia, Shell
Simulation & Modeling: High-fidelity physics simulation (multiscale/multiphysics, kinetic plasma modeling, radiative processes (synchrotron, inverse Compton), radiative transfer, relativistic MHD, hydrodynamics (SPH), stochastic particle dynamics, Lagrangian particle dispersion), numerical PDEs (Fokker–Planck, advection–diffusion), finite difference/volume methods, numerical linear algebra, model validation, physics-informed synthetic data generation, uncertainty quantification, parameter estimation
ML & Data: Scientific ML, language-model evaluation (scientific reasoning), surrogate modeling, Bayesian inference, Gaussian process regression, gradient-based optimization, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, neural networks, SVM, random forest, JAX (introductory coursework)
HPC & GPU: OpenMP, MPI, OpenACC (GPU exploration), SLURM, PBS, HDF5, large-scale parallel computing (Frontera)
Geospatial: QGIS, raster/vector processing, Sentinel-1/2 multispectral data, DEM terrain analysis, terrain and elevation derivatives, geospatial feature engineering for ML
Tools: Git and code review, Docker, Azure, CI/CD (Jenkins, OpenShift), Kafka, Splunk, Jira, QGIS, ParaView, VisIt

Experience

Independent Contractor Jan 2026 – Present
Mercor Remote
Physics subject-matter expert evaluating frontier language models for multiple leading AI labs.

- Review 50+ expert-level physics, hydrodynamics, astrophysics, and cosmology problems for leading AI labs; write original prompts that stump frontier models and co-develop others with fellow writers.
- Diagnose failure modes in model reasoning and write structured feedback to improve scientific accuracy.
- Benchmark model performance and quality-check solutions to complex STEM problems.

Spatial Data Scientist May 2025 – Jan 2026
TealWaters Seattle, WA
Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.
- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Engineered features from Sentinel-1/2 multispectral imagery and terrain derivatives for ML land-cover and wetland models; built Python tools and QGIS visualizations.
- Presented results to managers and decision-makers, trained 2 colleagues on the WIP tool, and worked with software engineers to move prototype tools into production.

Independent Researcher & Open-Source Developer Jan 2024 – May 2025
Self-directed Seattle, WA
Continued simulation research and open-source development between industry roles.

- Co-developed Tleco, an open-source Rust/Python toolkit that solves the Fokker–Planck equation coupled to radiative transfer to model radiation from relativistic outflows (ApJ 2024).
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library that simulates wind-driven spore dispersal over forested, hilly terrain, parallelized with RcppParallel and OpenMP.
- Used AI coding assistants in the most recent stages of development; changes were accepted only if they passed the unit-test suite and, for some functions, matched reference solutions.
- Invited talk on machine learning in environmental science at the University of Washington (2024).

Software Engineer Apr 2022 – Jan 2024
Paychex Remote
Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Built Java Kafka consumers that moved data between enterprise databases, deployed through Jenkins and OpenShift CI/CD pipelines.
- Created Splunk dashboards tracking the 200–400 users shared between enterprise systems and Oracle EBS, with alerts that flagged production incidents.
- Prepared, validated, and analyzed Oracle EBS data with SQL.
- Wrote full test suites for 5+ Python services in the CI pipeline, keeping production defect rates low.

Postdoctoral Research Associate Feb 2021 – Apr 2022
Rochester Institute of Technology Rochester, NY
HPC simulations of supermassive black hole binaries in an NSF-funded project.

- Led a 7-person team upgrading PatchworkMHD, a C code for magnetohydrodynamic simulations of supermassive black hole binaries.
- Added black hole spin to the simulations with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 10,000 node-hours, and identified performance bottlenecks in the parallel code.
- Calibrated Paramo parameters against Fermi-LAT gamma-ray spectra with gradient-based optimization, comparing simulation output with data and updating the model automatically; supervised a graduate student on the same problem.
- Co-authored 2 papers from a multi-institution study of binary neutron star mergers and a third on blazar heating and cooling.

Postdoctoral Research Fellow Oct 2018 – Nov 2020
Purdue University West Lafayette, IN
Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Designed and wrote Paramo, an open-source Fortran 95 library that solves the Fokker–Planck equation for relativistic particle distributions coupled to radiative transfer; optimized it with OpenMP and tested GPU offload with OpenACC.
- Paramo has been adopted by other research groups, including Combi & Siegel (2023).
- Served as co-I on two NASA Fermi Guest Investigator grant proposals; the funded one ($68K) supported the finding that blazar jets launch with similar energy per baryon, independently of their power.
- Added numerical-integration and OpenMP features to Paramo and built Python tools to compute energy losses and spectra of high-energy particles in gamma-ray burst afterglows.
- Mentored 3 graduate students (1 M.S., 2 Ph.D.), co-authoring papers with each.
- Wrote Python scripts for statistical modeling of the COVID-19 outbreak in Mexico with a multidisciplinary group, and co-wrote Spanish-language infographics and blog posts against misinformation.

Postdoctoral Research Fellow Jan 2018 – Sep 2018
Universidad Michoacana de San Nicolás de Hidalgo Morelia, Mexico

- Built a Python pipeline that turned GRTrans black hole simulation images into training data for an SVM classifier of black hole spin (grayscale conversion, intensity normalization, spatial feature extraction).
- Released an open-source Python tool for radiative-transfer spectra and light curves.
- Organized an HDF5 workshop to train graduate students in high-volume data storage.

Graduate Research Assistant Oct 2011 – Jul 2017
Universitat de València Valencia, Spain

- Automated runs of the C-SPEV radiative-transfer code with Shell and Python pipelines: 10–60 simulations per project and tens of GB of HDF5 output, with data-quality checks for downstream analysis.
- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.
- Analyzed multi-wavelength observations from the NASA Fermi-LAT telescope and the Very Long Baseline Array (VLBA), including cross-matching sources across catalogs by sky coordinates and fitting log-parabola spectra.
- Fit C-SPEV simulation models to telescope data, identifying spectra and light-curve patterns that quantify plasma magnetization in blazars.
- Extended C-SPEV to compute discrete and continuous spectra from particle distributions of arbitrary shape, without affecting simulation runtime.
- Published 2 first-author papers in MNRAS.

Education

Ph.D. in Physics Oct 2011 – Jul 2017
Universitat de València Valencia, Spain
Excellent *cum laude*
Advisors: Prof. Miguel A. Aloy and Dr. Petar Mimica
Thesis: Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows

M.Sc. in Physics Aug 2009 – Sep 2011
Universidad Michoacana de San Nicolás de Hidalgo Morelia, Mexico
Advisor: Prof. José A. Cervera
Thesis: *Study of TOV stars with the SPH method* (wrote an SPH hydrodynamics code to study the stability of relativistic stars)

B.Sc. in Physics Aug 2004 – Dec 2008
Universidad Autónoma del Estado de México Toluca, Mexico
Advisor: Prof. Francisco S. Guzmán
Thesis: *Numerical solution of null geodesics for the generation of gravitational lenses produced by spherically-symmetric and static spacetimes*
Lic. Juan Josafat Pichardo Cruz Award

Software and Projects

WindsOfChange Aug 2024 – Present
Lagrangian particle-dispersion library (C++/R) github.com/UW-ForestMycobiomeLab/WindSporesHills

- Translated an existing MATLAB code to R and C++, producing a library for Lagrangian particle dispersion over hilly terrain with forest canopy, driven by prescribed wind fields, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004). Developed with UW collaborators; manuscript in preparation.
- Parallelized particle tracking with RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.
- Used AI coding assistants in the most recent stages of development; changes were accepted only if they passed the unit-test suite and, for some functions, matched reference solutions.

Blazar Population ML Jan 2026 – Present
Physics-informed ML (Python, scikit-learn, PyTorch), developed with Claude github.com/altjerue/ilwikak-ml

- Applying classical ML (PCA, UMAP, Gaussian mixture models, HDBSCAN) to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to test whether physical jet properties can be recovered from multi-wavelength observables; documented missingness and selection effects feature by feature.
- Fitting Gaussian process regression to a lower bound on the jet Lorentz factor for a 334-source MOJAVE cross-matched subsample.
- Building a neural-network surrogate of the forward model (jet parameters to observable SEDs), to be benchmarked against Tleco simulations, to enable fast parameter inference.
- Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves (in progress).

Tleco Jan 2024 – Sep 2024
Radiative-transfer toolkit (Rust/Python) github.com/zkdavis/Tleco

- Numerical PDE solvers, particle-distribution algorithms, and spectral calculations for relativistic plasmas; successor to Paramo.

Paramo Oct 2018 – Apr 2022
Radiative-transfer library (Fortran 95, OpenMP) github.com/altjerue/paramo

- Fokker–Planck and radiative-transfer solver for relativistic particle distributions, parallelized with OpenMP; used by other groups, including Combi & Siegel (2023).

PatchworkMHD Feb 2021 – Apr 2022
Large-scale MHD simulation code (NSF project)

- Led a 7-person upgrade of this production C code for HPC environments, adding black hole spin with under 5% runtime overhead.

Selected Publications (9 published articles, 3 first-author; 3 in preparation)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, MNRAS 513, 5766–5779 (2022), arXiv:2201.07790.
- **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.
- **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.
- Guzmán, F. S., & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, Phys. Rev. D 80, 084023 (2009), arXiv:1009.1250.